HakaseAI BioDigital Twin

# Layer 1 — In Vitro Digital Twin

## Technical Reference & Evaluation Methodology

How every Layer 1 feature works, what tools and data sources drive it, how the GO / WATCH / REVIEW / NO-GO verdict is computed, and an honest account of how far the platform can be trusted for use by practising scientists.

> **Intended use — read this first**
> HakaseAI Layer 1 is a **directional decision-support and triage tool** (“HAIOps”), not a laboratory assay or a regulatory instrument. It is built to **prioritise, rank, and flag** small-molecule candidates and to make its reasoning fully auditable. It is **not** validated as a substitute for wet-lab measurement, GLP safety studies, or any regulatory submission, and no verdict should be the sole basis for a go/no-go decision. Confidence, applicability domain, and provenance are surfaced on every output precisely so a scientist can judge when to trust a number and when to confirm it.

Document version 1.0 • Generated 19 June 2026 • Reflects the calibrated scorecard, calibration/conformal layer, and applicability-domain gating in the current build.

# 1 How confident are we — and who should use this?

Short answer: Layer 1 is **trustworthy as a triage and hypothesis-generation layer for trained scientists**, and it is deliberately engineered to fail safe — it abstains ("REVIEW") rather than issue a confident verdict when it is extrapolating. It is **not** yet prospectively validated as an autonomous decision-maker, and it should be used by people who can interpret an ADMET prediction, a read-across estimate, and an applicability-domain caveat.

Three design choices underpin that confidence:

- **Every number carries its provenance and reliability.** A value is labelled as measured (experimental), model-predicted (with the model's published benchmark accuracy), or heuristic. Endpoints the underlying models are known to be poor at (e.g. volume of distribution, half-life) are flagged *unreliable* rather than shown as fact.
- **The model never vetoes outside its competence.** A development-stopping NO-GO from a machine-learning prediction is only issued when the molecule is inside the model's applicability domain and the prediction is calibration-reliable; otherwise the candidate is routed to REVIEW for an orthogonal assay. This is the same principle the OECD and FDA expect of in-silico models.
- **Known chemistry bypasses the models.** Well-characterised drugs resolve to their published measured profiles (lookup-first), so the platform does not "re-guess" established facts.

> **What is and isn't validated**
>
> **Validated (retrospective, internal):** the hard-veto thresholds were calibrated on a DrugCentral 2024 approved set (n=152) plus a withdrawn-liability set (n=18), achieving a false-positive rate of 0.04 on approved drugs and 0.94 recall on withdrawn drugs (Youden's J = 0.90). The underlying ADMET models report TDC-benchmark AUROC of 0.84–0.93 on the safety endpoints.
>
> **Not yet established:** prospective validation against new wet-lab outcomes; a fitted (Platt/isotonic) probability calibration on a held-out split (the layer currently runs in reliability-weighting mode using the real published AUROCs); and any regulatory qualification. Read-across quality depends on ChEMBL coverage for the chemotype.

# 2 Layer 1 at a glance

Layer 1 takes a single small molecule (SMILES) and an optional biological target, and returns a calibrated triage verdict with a full evidence trail. The stages run as follows:

| # | Stage | What it produces |
|---|---|---|
| **1** | Structure intake & validation (RDKit) | Canonical structure, descriptors, validity check |
| **2** | Identification & novelty | Is this a known compound? PubChem/UniChem/Europe PMC consensus |
| **3** | Known-drug lookup-first | If matched → published measured profile (bypasses models) |
| **4** | ADMET prediction | 41-endpoint ADMET-AI profile + specialist hERG/DILI models |
| **5** | In-vitro digital twin | Potency, permeability, metabolic stability, PPB, selectivity, TI |
| **6** | Read-across | Anchors novel-compound estimates to measured ChEMBL analogues |
| **7** | Structural-alert scan | Deterministic reactive/genotoxic substructure detection |
| **8** | Calibration + AD gating | Turns raw probabilities into reliability-gated, vetoable signals |
| **9** | Benefit-risk context | FDA openFDA, Pharos target level, DailyMed regulatory cross-ref |
| **10** | Verdict evaluation | MPO composite + hard-veto gates → GO / WATCH / REVIEW / NO-GO |

Stages 4–9 each emit dimensions, flags, and provenance; stage 10 aggregates them. The remainder of this document describes each feature and then the verdict logic in full.

# 3 Features — mechanics, tools and logic

## 3.1 Structure intake & novelty identification

The input SMILES is parsed and validated with **RDKit**; molecular descriptors (MW, logP, H-bond donors/acceptors, TPSA, rotatable bonds, QED, synthetic-accessibility score) are computed here. Novelty is then resolved by a **multi-database consensus** rather than a single lookup:

- **PubChem** — exact-structure / CID resolution (is the compound catalogued?).
- **UniChem (EBI)** — cross-references the structure across >40 chemistry sources.
- **Europe PMC** — literature footprint; a structure discussed in papers/patents is not novel.

The resolver returns a 0–1 novelty score and a verdict (Known / Novel), and flags patent presence. This feeds the confidence model: a structure with no public footprint cannot carry a high-confidence label.

## 3.2 Known-drug lookup-first short-circuit

Before any model runs, the compound is matched against a curated known-drug set by InChIKey and Tanimoto similarity. On a match, Layer 1 returns the drug's **published measured profile** directly and skips the ML stack entirely. This is the single most important guard against the classic failure mode where stacked model thresholds reject every approved drug. A matched drug with a documented clinical liability (e.g. terfenadine's hERG block) is shown as a distinct **KNOWN-LIABILITY NO-GO** — visibly different from a model-driven rejection, because it reflects literature, not a prediction.

## 3.3 ADMET property prediction (ADMET-AI)

Novel chemistry is profiled with **ADMET-AI** (Swanson et al., *Bioinformatics* 2024) — a Chemprop directed-message-passing graph neural network with RDKit features, trained on the Therapeutics Data Commons (TDC) ADMET benchmark (41 endpoints). The platform transcribes the model's own published per-endpoint accuracy so each number is shown with its correct unit and an honest reliability tier. Key safety/ADME endpoints:

| Endpoint | Type | Unit | Benchmark | Reliability |
|---|---|---|---|---|
| **Human intestinal absorption (HIA)** | class | prob | AUROC 0.99 | high |
| **P-gp inhibition** | class | prob | AUROC 0.95 | high |
| **Aqueous solubility** | regr | log(mol/L) | $R^2$ 0.82 | high |
| **BBB penetration** | class | prob | AUROC 0.90 | high |
| **CYP3A4 / 2C9 / 2C19 inhibition** | class | prob | AUROC 0.91 | high |
| **hERG blockade** | class | prob | AUROC 0.84 | moderate |
| **Ames mutagenicity** | class | prob | AUROC 0.88 | high |
| **DILI (liver injury)** | class | prob | AUROC 0.88 | high |
| **Clinical-trial toxicity (ClinTox)** | class | prob | AUROC 0.93 | high |
| **Carcinogenicity** | class | prob | AUROC 0.77 | moderate |
| **Volume of distribution (VDss)** | regr | L/kg | $R^2$ –1.21 | unreliable |
| **Half-life** | regr | hr | $R^2$ –2.39 | unreliable |

The two negative-$R^2$ endpoints are worse than predicting the dataset mean; the platform explicitly marks them unreliable rather than presenting the number as fact. This honesty about model limits is a deliberate trust feature.

## 3.4 Specialist safety models

Where a dedicated model outperforms the general ADMET-AI head, it takes precedence: a **CardioTox-ML hERG** model is preferred over the ADMET-AI hERG output, and a **DILI-ML** (DILIrank-trained) model is preferred for hepatotoxicity. The overlay records which model supplied each probability so the verdict's ML coverage can be reported.

## 3.5 In-vitro digital-twin panel

Alongside ADMET, Layer 1 simulates the readouts of a standard in-vitro panel. Each axis is tagged ML or heuristic, and a heuristic value never masquerades as a measurement:

- **Potency / on-target IC50** (nM) with a potency class.
- **Permeability** — Caco-2 class.
- **Metabolic stability** — microsomal class.
- **Plasma-protein binding** — bound-fraction class.
- **hERG risk** — physico-chemical heuristic band used only when no ML/patch-clamp value exists.
- **Selectivity** — in-silico promiscuity score and class.
- **Cell viability / therapeutic index** — CC50 vs IC50 safety window.

## 3.6 Read-across engine (the novel-compound anchor)

For a genuinely novel molecule a pure model prediction is an extrapolation. Read-across is the regulator-accepted way to get closer to the truth without a lab (OECD 2014 grouping guidance; ECHA RAAF 2017): find the structurally nearest compounds that *do* have a measured value and anchor the estimate to them. The engine queries **EBI ChEMBL** with a Tanimoto similarity ladder (70% → 55% → 40%), pulls measured activities (hERG = CHEMBL240, on-target = selected target), and computes a similarity-weighted estimate in the log domain:

```
estimate = 10 ^ ( Σ w·log10(value) / Σ w ),   w = similarity²,   with a geometric-SD fold spread and an applicability-domain confidence (High / Moderate / Low / None) from nearest-neighbour Tanimoto and neighbour count.
```

When no measured neighbour exists, the endpoint is reported as a pure extrapolation (lower confidence) — never a fabricated number. This same applicability-domain signal drives the veto gate (section 5).

## 3.7 Structural-alert scan

A deterministic SMARTS scan flags reactive and genotoxic substructures, drawn from five published libraries: **Kazius** (Ames toxicophores), **Brenk**, **PAINS** (Baell & Holloway), **NIH-MLSMR**, and **BMS**. Alerts are tiered:

- **Reject (deterministic veto):** genuine reactive/genotoxic chemistry — nitroaromatics, nitrosamines, epoxides, acyl/sulfonyl halides, Michael-acceptor warheads, peroxides, heavy metals. These are chemistry facts and may stop a program regardless of the ML scores (ICH M7).
- **Warn (confidence modifier):** everything whose basis is assay-interference or weaker association. Per the "Phantom PAINS" analysis (Capuzzi et al. 2017 — 87 FDA-approved drugs contain a PAINS alert), **no PAINS-category alert is ever allowed to hard-veto**; it only lowers confidence and prompts orthogonal confirmation.

> **Why this matters for false positives**
>
> A previous build flagged a safe tocopherol-class antioxidant as toxic because a hydroxyl-vs-ether SMARTS over-matched. The fix was twofold: tighten the SMARTS to require true hydroxyls, and enforce the policy that assay-interference alerts cannot stop development on their own. Reactive/genotoxic chemistry still vetoes; promiscuity-class patterns do not.

## 3.8 Calibration & conformal-reliability layer

This is the layer that decides whether a model prediction is trustworthy enough to act on. A raw graph-neural-net output is a score, not a calibrated probability, and its discrimination is bounded by the model's own benchmark AUROC. Each safety probability is therefore wrapped in a reliability assessment:

```
reliability = AUROC × (0.5 + 0.5×margin) × AD_factor   [margin = |p − 0.5|×2; AD_factor:
High 1.0, Moderate 0.82, Low 0.55, None 0.35]
interval = [ p − (1−reliability)×0.35 ,  p + (1−reliability)×0.35 ]
```

The decision rule borrows from **conformal prediction** (Vovk; Angelopoulos & Bates 2021; conformalised graph learning for ADMET, JCIM 2024): a hard veto is upheld **only if the lower confidence bound still exceeds the hard threshold**. If the interval straddles the threshold, or the molecule is out of domain, or measured analogues contradict the call, the liability is recorded as an **escalate** signal instead of a stop. An optional fitted Platt/isotonic table is supported; absent one, the layer reliability-weights using the real AUROC and says so — it never fabricates a calibration curve.

## 3.9 Applicability-domain gating

OECD (Q)SAR validation principle 3: predictions outside the training domain are extrapolations and "less likely to be reliable." The molecule's domain confidence is taken from the read-across nearest-neighbour support (in-domain Tanimoto ≥ 0.65, border ≥ 0.40, out-of-domain below). Out-of-domain ML signals are demoted from veto to escalate — an extrapolating model may not stop a program.

## 3.10 Benefit-risk & regulatory context

For context (not as a veto), Layer 1 layers in FDA-style benefit-risk framing using **openFDA** drug-label data (boxed-warning / label signal), **NCATS Pharos** target development level (Tclin/Tchem/Tbio/Tdark — how validated the target is), and **DailyMed** (FDA SPL labels) as a regulatory cross-reference for known drugs. Disease severity and unmet need scale the risk tolerance, mirroring the FDA Benefit-Risk Framework (2023). The safety veto remains sovereign: benefit context can never upgrade a molecule past a genuine safety stop.

# 4 The scoring math: MPO desirability composite

Candidates are ranked the way a medicinal-chemistry portfolio committee triages: each endpoint is mapped to a desirability $d \in [0,1]$ (Harrington 1965; Derringer & Suich 1980) and the axes are combined as a **weighted geometric mean** — the same scalarisation behind QED (Bickerton 2012) and CNS-MPO (Wager 2010):

$$D = \exp\left( \Sigma_i \, w_i \cdot \ln(d_i) \,/\, \Sigma_i \, w_i \right) \times 100$$

The geometric mean is the point: a single weak axis (a strong DILI signal, say) drags the whole composite down — you cannot “average away” a safety liability with good permeability, which is exactly how committees behave and removes the additive-scoring failure mode. Weights are renormalised over only the dimensions that could actually be computed (a missing endpoint is dropped, never defaulted to a flattering value, and that absence lowers confidence). Liability probabilities use a Wager-style piecewise-linear desirability: d = 1 below the watch threshold, declining linearly to d_min = 0.05 at the hard threshold.

Intended MPO weights (renormalised per molecule; safety aggregates to ~0.44, reflecting that safety and PK are the leading causes of clinical attrition, Waring 2015):

| Axis | Dimension | Weight |
|---|---|---|
| **Efficacy** | Potency (on-target) | 0.12 |
| **ADME** | Absorption | 0.10 |
| **ADME** | Metabolic stability | 0.10 |
| **ADME** | Distribution | 0.06 |
| **Safety** | Cardiac (hERG) | 0.14 |
| **Safety** | Hepatic (DILI) | 0.14 |
| **Safety** | Genotoxicity (Ames) | 0.10 |
| **Safety** | Other tox (ClinTox/Carcinogen) | 0.06 |
| **Developability** | Selectivity | 0.08 |
| **Developability** | Drug-likeness (QED/Ro5/SA) | 0.10 |

# 5 Hard-veto gates (regulatory stage-gates)

Hard vetoes are **not** averaged into the composite — they are deterministic stage-gates grounded in ICH guidance. They fall into two classes. **Deterministic** vetoes (reactive/genotoxic structural rejects; ≥3 Lipinski violations) are chemistry facts and fire regardless of applicability domain. **Model-derived** vetoes (hERG, DILI, Ames, ClinTox, carcinogenicity) must pass the calibration/AD gate of section 3.8 before they can stop a program.

Calibrated thresholds (re-fitted so the union false-positive rate on approved drugs is ≤5% while recall on withdrawn-liability drugs stays ≥90%; these apply to *novel* chemistry only — known drugs bypass them via lookup-first):

| Liability | Watch | Hard veto | Regulatory basis |
|---|---|---|---|
| **hERG block** | 0.55 | 0.92 | ICH S7B / E14 |
| **DILI** | 0.55 | 0.88 | leading attrition cause |
| **Ames mutagenicity** | 0.55 | 0.85 | ICH M7 / S2(R1) |
| **Clinical toxicity** | 0.60 | 0.88 | ClinTox |

| Liability | Watch | Hard veto | Regulatory basis |
|---|---|---|---|
| **Carcinogenicity** | 0.60 | 0.88 | ICH S1 / S2(R1) |
| **Lipinski Ro5 violations** | 2 | ≥3 | Lipinski 2001 |

# 6 How the final verdict is evaluated

The scorecard resolves a verdict by walking an ordered decision pipeline. The order matters: measured evidence and deterministic chemistry are considered before, and override, probabilistic model signals.

| Step | Check | Outcome |
|---|---|---|
| **A** | Known-drug lookup matched? | Yes → return measured profile: clean → GO; label monitor → WATCH; documented liability/withdrawn → KNOWN-LIABILITY NO-GO. Stop. |
| **B** | Compute desirability dimensions + MPO composite | Each axis scored; weighted geometric-mean composite (0–100). |
| **C** | Deterministic veto present? (structural reject / ≥3 Ro5) | Yes → NO-GO regardless of AD or ML. |
| **D** | Model-derived liability over hard threshold? | Route through calibration + AD gate (assessVeto). |
| **D1** | • in-domain, reliable, lower-bound > threshold, no contradiction | Veto upheld → NO-GO. |
| **D2** | • out-of-domain / unreliable / interval straddles / contradicted | Escalate signal → REVIEW (not NO-GO). |
| **E** | No vetoes: warn flags or composite below floor? | Yes → WATCH. |
| **F** | Composite ≥ GO floor (67) and all guards pass? | Yes → GO; else WATCH. |
| **G** | Conservative GO guard | GO downgraded to WATCH unless confidence High, ML safety coverage ≥66%, in-domain, ≥6 dimensions, and composite margin above floor. |

## 6.1 Verdict classes

| Verdict | Score band | Meaning |
|---|---|---|
| **GO** | 70–100 | Favourable profile, no hard or escalate liabilities; advance to Layer 2. |
| **WATCH** | 55–69 | Advance with monitoring; minor flags or composite below GO floor. |
| **REVIEW (escalate)** | 40–54 | Strong predicted liability the model could not confirm (out-of-domain / not reliable / contradicted). Abstain; run an orthogonal assay. |
| **NO-GO** | 1–39 | Deterministic veto or a calibration-reliable in-domain model veto. |
| **KNOWN-LIABILITY NO-GO** | ~15–38 | Matched a known drug with a documented clinical liability — published evidence, not a model rejection (distinct UI colour). |

The triage score is banded so it can never disagree with the verdict, guaranteeing a correct sort when ranking a whole library. Confidence (High/Moderate/Low) is reported separately, driven by ML coverage of the safety axes, data completeness, and measured-analogue support.

## 6.2 Worked example — why a safe antioxidant no longer fails

A tocopherol/chromanol-type antioxidant returns a high model DILI probability but has **no close measured analogues** in ChEMBL (out-of-domain). Old behaviour: the raw probability crossed the hard threshold → NO-GO (a false stop). New behaviour: step D routes the signal through the calibration gate; out-of-domain → reliability below the veto bar → step D2 → **REVIEW (score ~48)** with the explicit reason “strong predicted liability not confirmable in-silico; escalate for an orthogonal assay.” The platform abstains rather than issue a confident wrong verdict — the behaviour a scientist should expect.

# 7 Evaluation status & recommended use

| Aspect | Status |
|---|---|
| **Hard-veto threshold calibration** | Retrospective, internal (DrugCentral n=152 approved + 18 withdrawn): FPR 0.04, recall 0.94, Youden J 0.90. |
| **ADMET model accuracy** | Published TDC benchmark AUROC 0.84–0.93 (safety); two PK regression endpoints flagged unreliable. |
| **Probability calibration** | Reliability-weighting using real AUROC + applicability domain (functional); fitted Platt/isotonic table optional, not yet generated. |
| **Applicability-domain gating** | Active — out-of-domain ML signals cannot hard-veto. |
| **Continuous validation** | Feedback log live (reviewer corrections persisted); regression set accumulates for ongoing threshold/calibration checks. |
| **Prospective wet-lab validation** | Not yet performed. |
| **Regulatory qualification** | None — not for submission use. |

**Appropriate use**

**Use Layer 1 to:** triage and rank libraries; prioritise which compounds and which assays to run first; surface and explain liabilities; generate hypotheses; and document an auditable rationale. Treat GO as “promising, confirm in Layer 2/lab,” WATCH as “monitor,” REVIEW as “the model is unsure — get data,” and NO-GO as “strong reason to deprioritise or redesign.”

**Not appropriate**

**Do not** use any Layer 1 verdict as the sole basis for a go/no-go decision, as a substitute for required wet-lab or GLP studies, or in any regulatory submission. Predictions on out-of-domain chemistry are explicitly flagged REVIEW for exactly this reason.

# 8 Design principle: provenance & honesty

Every sub-score and flag carries the source it came from (measured / ML / heuristic) and nothing is invented: when an endpoint cannot be computed the dimension is dropped rather than defaulted, and that absence lowers confidence. The platform is governed as “directional decision-support, not a lab replacement” (HAIOps), and is built so a trained scientist can always see *why* a verdict was reached and decide whether to trust or confirm it.

# 9 Key references


- Swanson K. et al. (2024) ADMET-AI. *Bioinformatics* 40:btae416.
- Bickerton G.R. et al. (2012) Quantifying the chemical beauty of drugs (QED). *Nat. Chem.* 4:90.
- Wager T.T. et al. (2010) CNS multiparameter optimization. *ACS Chem. Neurosci.* 1:435.
- Harrington E.C. (1965) The desirability function. *Ind. Qual. Control* 21:494; Derringer & Suich (1980) *J. Qual. Technol.* 12:214.
- Kazius J. et al. (2005) Derivation of toxicophores. *J. Med. Chem.* 48:312; Brenk R. et al. (2008) *ChemMedChem* 3:535.
- Baell J. & Holloway G. (2010) New substructure filters (PAINS). *J. Med. Chem.* 53:2719; Capuzzi S. et al. (2017) Phantom PAINS. *JCIM* 57:417.
- OECD (2014) Guidance on grouping of chemicals (read-across); ECHA Read-Across Assessment Framework (2017); OECD (Q)SAR validation principles.
- Angelopoulos A. & Bates S. (2021) A gentle introduction to conformal prediction. arXiv:2107.07511; conformalised graph learning for ADMET, *JCIM* 2024.
- ICH M7(R2), S7B, E14, S2(R1), S1; Lipinski C. et al. (2001) *Adv. Drug Deliv. Rev.* 46:3.

- FDA (2025) Considerations for the Use of AI to Support Regulatory Decision-Making for Drug and Biological Products (draft).
- Waring M.J. et al. (2015) Attrition of drug candidates. *Nat. Rev. Drug Discov.* 14:475.